

Randomized Numerical Linear Algebra for the Sciences

Tyler Chen (陈泰乐)

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<https://research.chen.pw/talks/SHTech26.pdf>

Overview

History:

- PhD in Applied Math at University of Washington
- Assistant Professor / Courant Instructor at Tandon CS and Courant Math
- Applied Research Lead at JPMorganChase

Research: I design and analyze fast and theoretically justified (randomized) algorithms for core linear algebra tasks.

Goal: My broad goal is to develop tools to support the advancement of knowledge in current scientific applications.

Misc: I really like working with students / teaching.

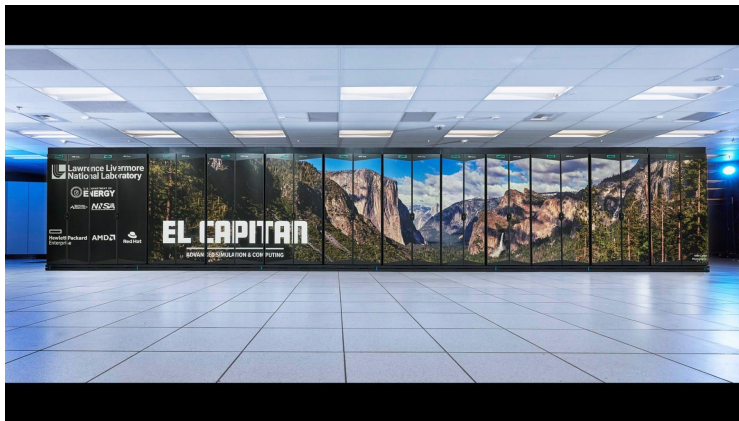
Numerical Linear Algebra

NLA is a core part of computational math.

Centered on designing/understanding algorithms to efficiently and accurately do core linear-algebra tasks:

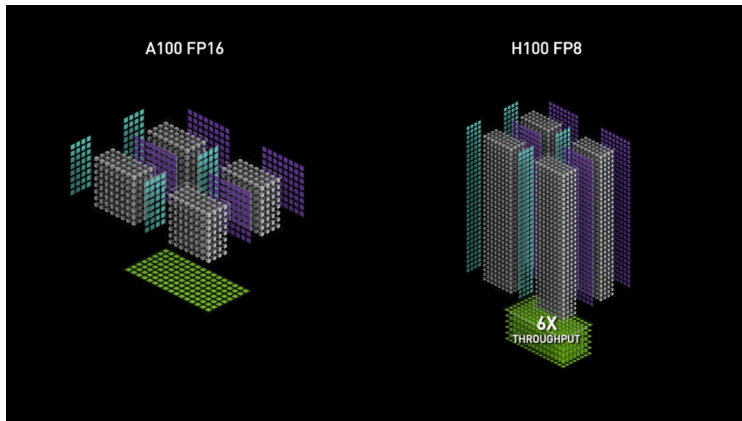
- compute $\mathbf{AB} + \mathbf{C}$
- factor $\mathbf{A} = \mathbf{LU}$ or $\mathbf{A} = \mathbf{QR}$
- solve $\mathbf{Ax} = \mathbf{b}$ or $\min_{\mathbf{x}} \|\mathbf{b} - \mathbf{Ax}\|$
- compute eigenvalues/SVD
- compute $f(\mathbf{A})$ or $f(\mathbf{A})\mathbf{b}$
- compute $\log(\det(\mathbf{A}))$
- etc.

Linear-algebra drives modern computing



43k CPUs + 43k GPUs + fast interconnects → simulate physics (using NLA)

Linear-algebra drives modern computing



NVidia makes devices to do matrix-operations fast

Numerical Linear Algebra across computing

Machine Learning

- Training/optimization
- Inference
- Model compression
- Explainability

Scientific Computing

- PDE solvers
- Quantum simulations
- Climate modeling
- Molecular dynamics

Data Science

- PCA/clustering/regression
- Compression
- Netflix problem
- Page rank

Finance

- Portfolio optimization
- Risk assessment
- Option pricing
- Covariance estimation

Randomized Numerical Linear Algebra

Recently, there has been a widespread paradigm shift: **randomization** is now a critical tool in algorithm design (e.g. Stochastic Gradient Descent enabled NN/LLMs)

RandNLA aims to develop randomized algorithms for linear algebra tasks.

- randomized algorithms have provided orders-of-magnitude speedups in computation, and the best theoretical running times
- the field's success is due to a joint effort of theoretical computer science and numerical analysis communities (regular joint workshops / events)

Representative work

Machine learning/data science

- Theoretical guarantees for algorithms widely used in ML
ICML 21 long presentation, IMA J. Num. Analysis 24, NeurIPS 25, ...
- Krylov subspace methods for low-rank approx/matrix functions
SIMAX 23, NeurIPS 24 spotlight, SISC 25, SODA 26, ...

Quantum Physics/Chemistry

- Algorithms for quantum thermodynamics
J. Comp. Phys. 22/24, SISC 24, ...

Operator Learning

- What can we learn about (structured) matrices from matrix-vector products?
SIMAX 25/26, SODA 25, COLT 26 submission, ...

Classical NLA

- Algorithms for HPC/GPU/etc.
SISC 21/22/24, PDSEC 26, ...
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SIMAX, SISC, ETNA, Num. Alg., ...

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Today's talk: operator learning

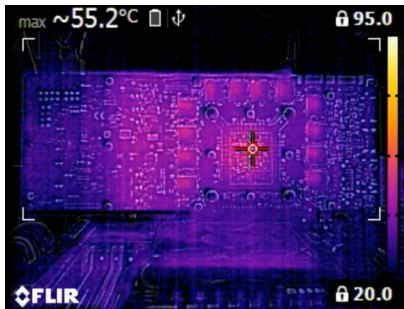
Today, I'll talk about a line of work on operator learning, which uses RandNLA to solve open problems in the emerging field of Scientific Machine Learning.

Operator learning aims to discover or approximate an unknown operator (map between functions), which is often the solution operator for a partial differential equation (PDE).

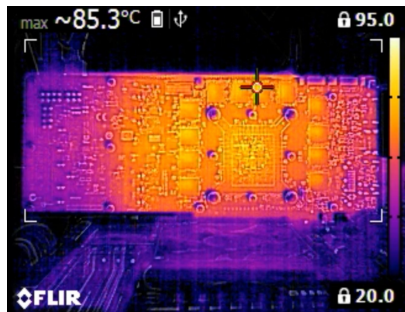
Note: I'll mostly be speaking about the theoretical side of things. There are also a very concrete applications of this line of work to speeding up real-world computation via preconditioning (ask me at the end!)

Operators describe physics

Physical processes often map a function f to a function u .



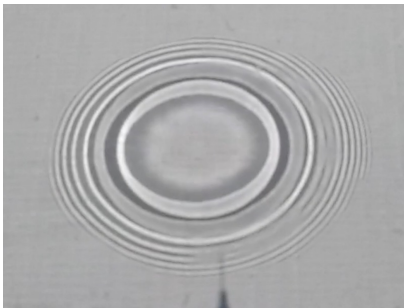
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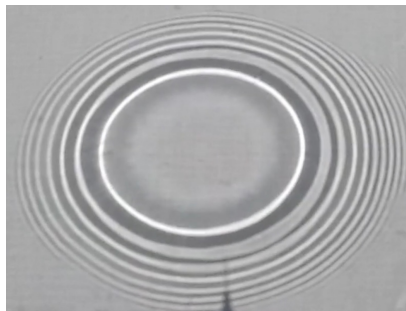
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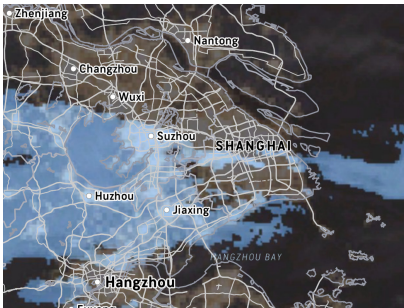
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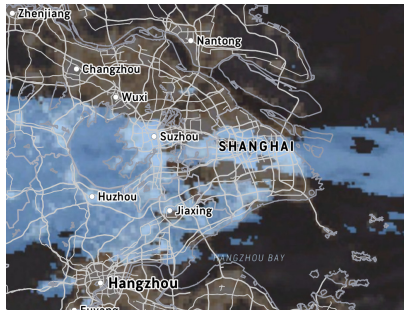
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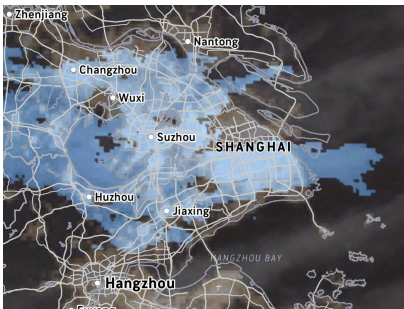
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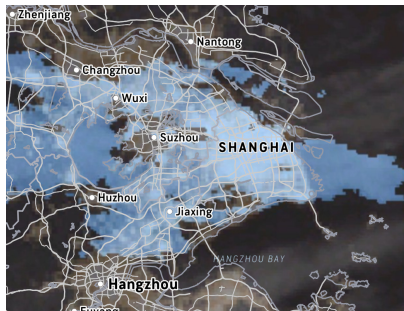
u

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f



u

Operator Learning

Setting: Define underlying operator taking functions f to functions u .

Goal: Learn about operator from observed input-output pairs: $(f_1, u_1), \dots, (f_m, u_m)$.

Things to think about:

- What types of operators can we learn?
- Do we need to be able to choose f_i ? What if it's randomly generated?
- How much data do we need?
- Are there limits on what can be learned?

Mapping operator learning to NLA

Q: How can we use our expertise in RandNLA to advance operator learning/SciML?

A: Distill to simple case: linear operator acting on finite dimensional spaces.

- Continuous functions become vectors and the operator becomes a matrix!
- Even this simple case is rich enough to capture a lot of relevant scientific problems and has proved hard to make progress in.

New goal: What can we learn about a matrix from a few matrix-vector products (matvecs) with $\mathbf{A}$?

- a few matvecs: $(\mathbf{x}_1, \mathbf{A}\mathbf{x}_1), \dots, (\mathbf{x}_m, \mathbf{A}\mathbf{x}_m)$
- $\mathbf{x}_i$ may or may not be chosen adaptively based on previous outputs

Matrix Approximation

Structured Matrix Approximation.

Fix some family S of matrices. How many matrix-vector products to (unknown, fixed, arbitrary) $\mathbf{A} \in \mathbb{R}^{n \times n}$ are required to learn (a parameterization of) $\tilde{\mathbf{A}} \in S$ such that

$$\|\mathbf{A} - \tilde{\mathbf{A}}\|_F \leq \Gamma \cdot \min_{\mathbf{X} \in S} \|\mathbf{A} - \mathbf{X}\|_F.$$

Interesting for $\Gamma = (1 + \varepsilon)$, $\Gamma = 2$, $\Gamma = \log(n)$, $\Gamma = n^{0.01}$, etc.

Past work: A lot of work on low-rank approximation.

My work: lots of scientifically relevant families!

- sparse matrices (SIMAX 2025)
- hierarchical matrices (SODA 2025, SIMAX 2026)
- general families (COLT 2026 submission)
- butterfly matrices (upcoming)

Warm up: Low-rank matrices

Let $S = \{\text{rank-}k \text{ matrices}\}$. Lots of work on this problem!¹

The Randomized SVD (RSVD) is a well-known algorithm for obtaining an approximation to $\mathbf{A}$ from S :

1. Sample Gaussian matrix $\mathbf{\Omega}$ with $m/2$ columns
2. Form $\mathbf{Q} = \text{orth}(\mathbf{A}\mathbf{\Omega})$
3. Compute $\mathbf{X} = \mathbf{Q}^\top \mathbf{A}$ (minimize: $\|\mathbf{A} - \mathbf{Q}\mathbf{X}\|_F$)
4. Output $\tilde{\mathbf{A}} = \mathbf{Q}[\mathbf{X}]_k$

Theorem. When $m = O(k/\varepsilon)$,

$$\|\mathbf{A} - \tilde{\mathbf{A}}\|_F \leq (1 + \varepsilon) \cdot \min_{\mathbf{X} \text{ rank-}k} \|\mathbf{A} - \mathbf{X}\|_F.$$

¹Sarlos 2006; Clarkson and Woodruff 2009; Halko, Martinsson, and Tropp 2011; Tropp and Webber 2023, etc.

Low-rank is understood, but not rich enough

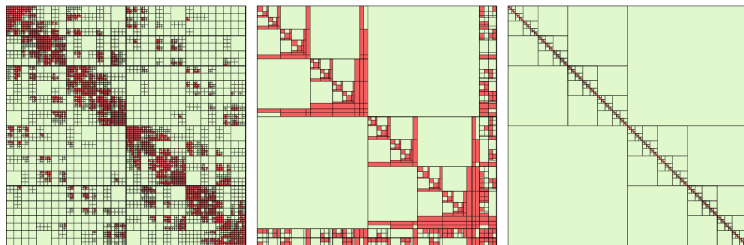
Decades of work mean we know a LOT about low-rank approximation, and have nearly-optimal algorithms.

However, matrices arising from scientific applications aren't always low-rank!!



Hierarchical matrices

Today, S will be some family of **hierarchical matrices**.



example classes: hierarchical off-diagonal low-rank (HODLR), hierarchical semi-separable (HSS), $\mathcal{H}^1$, $\mathcal{H}^2$, hierarchical off-diagonal butterfly, etc.

Motivation: better PDE learning

2024 SIAM Linear Algebra Best Paper Prize winner asks:²



Q: Can we efficiently obtain a near-optimal HODLR approximation from matrix-vector products?

Resolving this immediately gives better algorithms for learning **elliptic PDEs**.

Theorem (Chen, et al; SODA 2025). Yes!!!

²Boullé and Townsend 2022.

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HODLR matrices

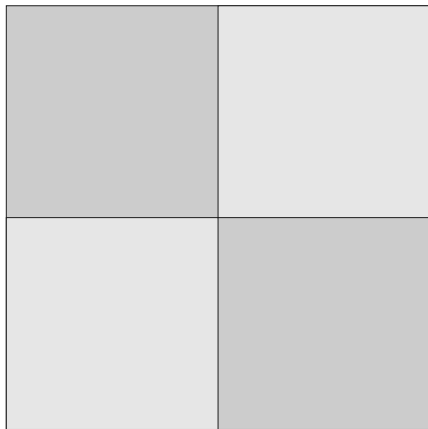


low-rank block



recursive block

HODLR matrices

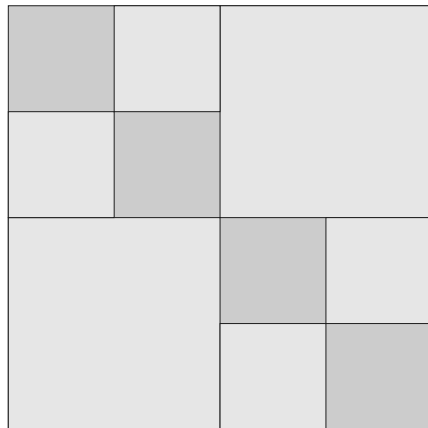


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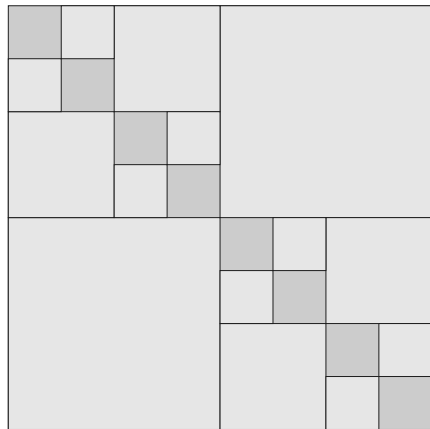


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HODLR matrices



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recursive block

Definition. Fix a rank parameter k . We say a $n \times n$ matrix $\mathbf{A}$ is HODLR(k) if $n \leq k$ or $\mathbf{A}$ can be partitioned into $(n/2) \times (n/2)$ blocks

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{1,1} & \mathbf{A}_{1,2} \\ \mathbf{A}_{2,1} & \mathbf{A}_{2,2} \end{bmatrix}$$

such that $\mathbf{A}_{1,2}$ and $\mathbf{A}_{2,1}$ are of rank at most k and $\mathbf{A}_{1,1}$ and $\mathbf{A}_{2,2}$ are each HODLR(k).

HODLR(k) matrices of size $n \times n$ have $O(kn \log(n))$ parameters. Moreover, given a HODLR factorization, lots of downstream linear algebra tasks can be accelerated.

Our main result

Theorem. There exist matvec algorithms which use $O(k \log(n)/\beta^3)$ products with $\mathbf{A}$ to obtain a $\text{HODLR}(k)$ matrix $\tilde{\mathbf{A}}$ satisfying

$$\|\mathbf{A} - \tilde{\mathbf{A}}\|_F \leq (1 + \beta)^{\log_2(n)} \min_{\mathbf{H} \in \text{HODLR}(k)} \|\mathbf{A} - \mathbf{H}\|_F.$$

Corollary. $(1 + \varepsilon)$ -optimal approximation with $O(k \log(n)^4/\varepsilon^3)$ matvecs

Corollary. $n^{0.01}$ -optimal approximation with $O(k \log(n))$ matvecs

Our main result

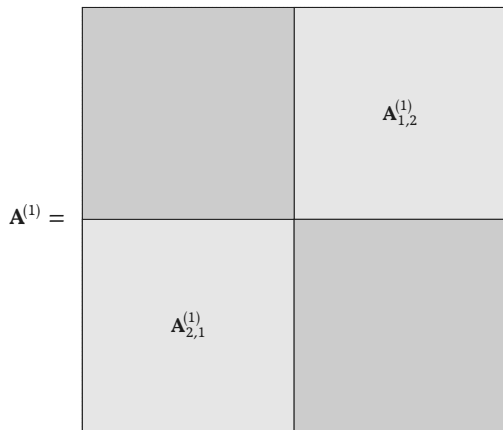
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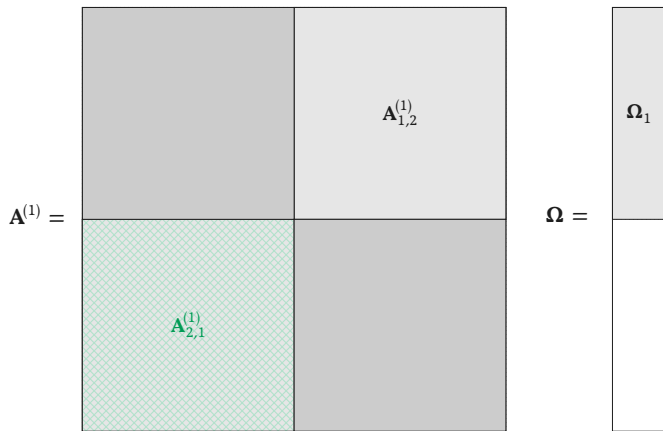
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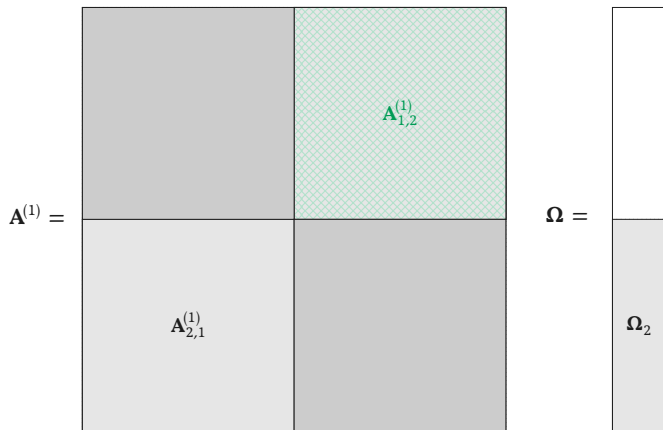
Our approach: robust peeling (idealized case, level 1)



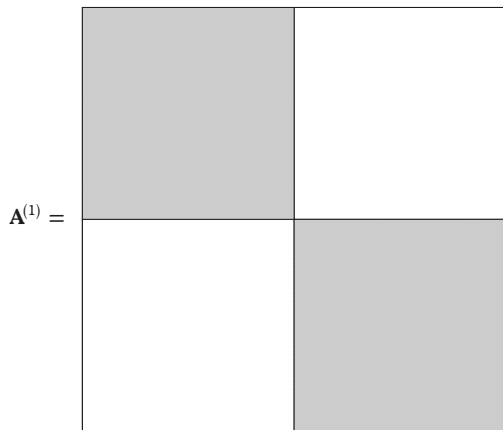
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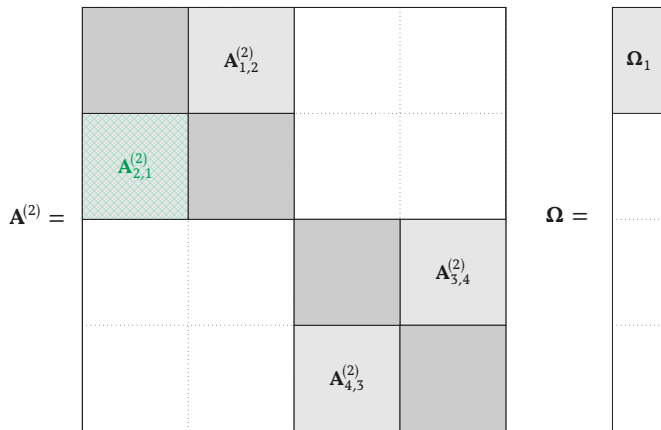
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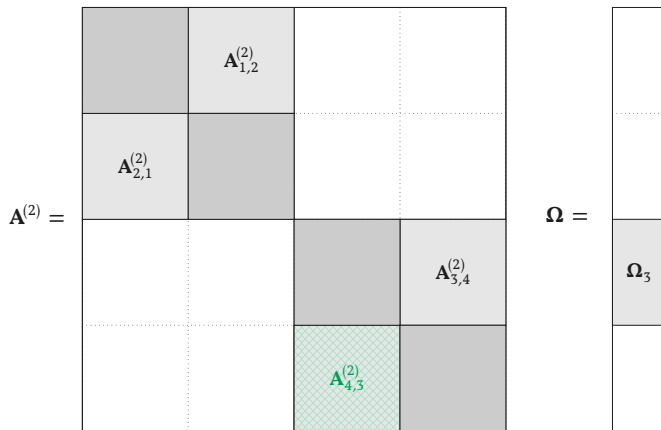
Our approach: robust peeling (idealized case, simultaneous approximation)

$$\mathbf{A}^{(2)} = \begin{array}{|c|c|c|c|} \hline \text{shaded} & \mathbf{A}_{1,2}^{(2)} & & \\ \hline \mathbf{A}_{2,1}^{(2)} & \text{shaded} & & \\ \hline & & \text{shaded} & \mathbf{A}_{3,4}^{(2)} \\ \hline & & \mathbf{A}_{4,3}^{(2)} & \text{shaded} \\ \hline \end{array}$$

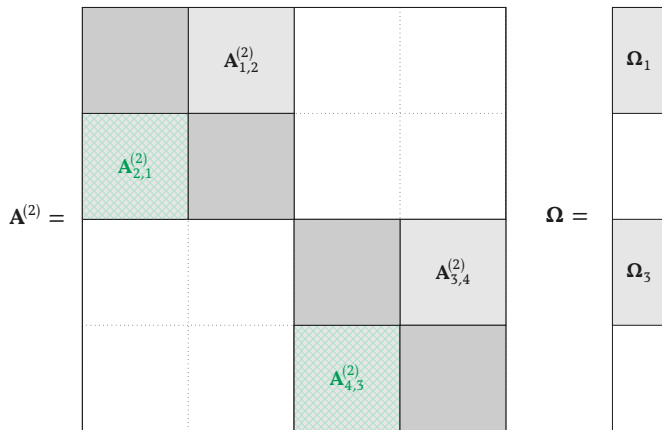
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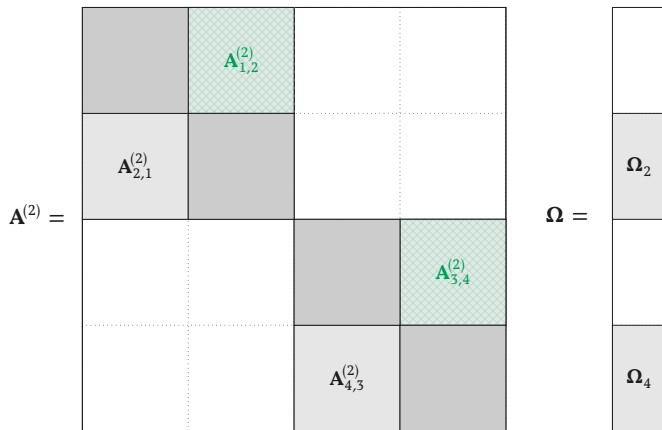
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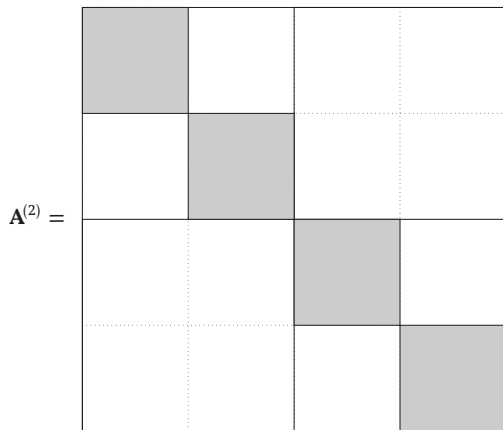
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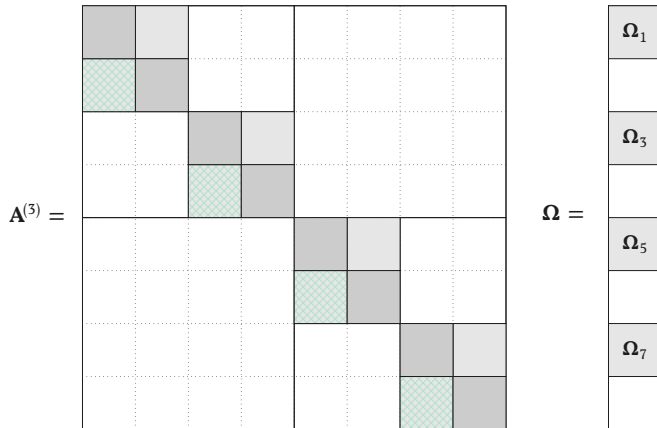
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What if there is error?



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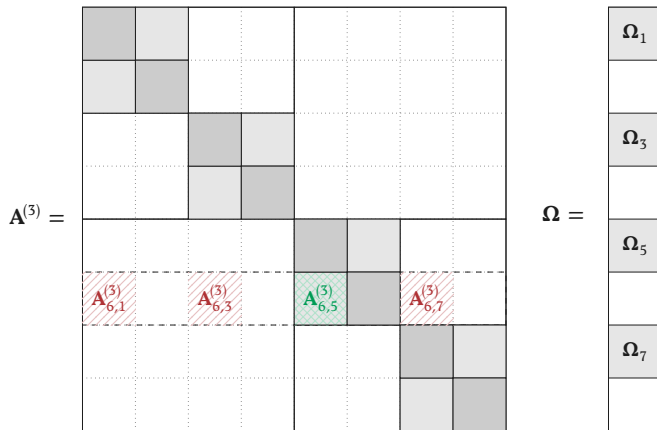
$$\mathbf{A}^{(3)} =$$

$\mathbf{A}_{6,5}^{(3)}$

$$\mathbf{\Omega} =$$

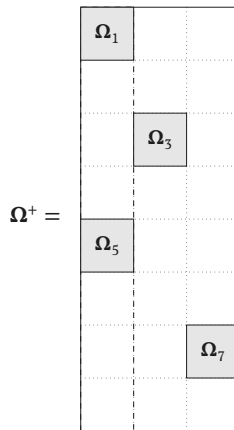
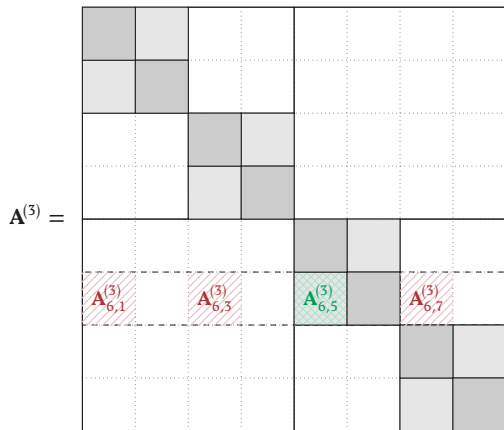
$\mathbf{A}_{6,5}^{(3)} \mathbf{\Omega}_5$

What if there is error?



$$\mathbf{A}_{6,5}^{(3)} \mathbf{\Omega}_5 + \mathbf{A}_{6,1}^{(3)} \mathbf{\Omega}_1 + \mathbf{A}_{6,3}^{(3)} \mathbf{\Omega}_3 + \mathbf{A}_{6,7}^{(3)} \mathbf{\Omega}_7$$

Perforated Block CountSketch



$$A_{6,5}^{(3)} \Omega_5 + A_{6,1}^{(3)} \Omega_1$$

Proof ingredient: A perturbation bound for the RSVD

Our algorithm is based on ideas like the RSVD and “perforated block count sketch”

- this required cross-disciplinary knowledge to identify and develop key algorithm tools.
- case-study in the use of RandNLA to solve problems in other fields

Our proof relies on technical ingredients including a new perturbation bound for the RSVD (which is likely of independent interest):

Theorem. Let $\mathbf{Q} = \text{orth}(\mathbf{B}\mathbf{\Omega} + \mathbf{E}_1)$ and $\mathbf{X} = \mathbf{Q}^\top \mathbf{B} + \mathbf{E}_2$. Then

$$\|\mathbf{B} - \mathbf{Q}[\mathbf{X}]_k\|_F \leq \underbrace{\|\mathbf{E}_1 \mathbf{\Omega}_{\text{top}}^\dagger\|_F + 2\|\mathbf{E}_2\|_F}_{\text{perturbations}} + \underbrace{\left(\|\mathbf{\Sigma}_{\text{bot}}\|_F^2 + \|\mathbf{\Sigma}_{\text{bot}} \mathbf{\Omega}_{\text{bot}} \mathbf{\Omega}_{\text{top}}^\dagger\|_F^2\right)^{1/2}}_{\text{classical RSVD bound}}.$$

Other interesting topics from this line of work

The matrix-vector query model often lets us prove **lower-bounds** against any matvec algorithm for a given task; i.e. study the **complexity** of a task!

Case study. For oscillatory operators, **butterfly matrices** are used. Similar to HODLR matrices, a butterfly matrix has $O(kn \log(n))$ parameters. However, all existing algorithms require $O(\sqrt{n})$ matvecs!! Lot of work to improve this rate..

We have shown a lower-bound of $\Omega(\sqrt{n})$ matvecs!!

- Maybe butterfly isn't the right class for matvec algorithms.

I'm also interested in developing a general theory for matrix-approximation (rather than class-specific algorithms).

- partial progress in Amsel, Avi, et al. 2025

In the coming years, I will work towards operationalizing RandNLA for widespread use across the sciences.

Operationalizing RandNLA

theoretical algorithm design/analysis

- I will expand the frontier of our understanding of (randomized) NLA algorithms for scientific applications.

problem identification

- My cross-disciplinary background makes me uniquely suited to identify connections between NLA and areas such as quantum physics/chemistry, operator learning, machine learning / data-science.

implementation oriented algorithms

- We must design algorithms with consideration of fine-grained cost tradeoffs relevant to implementation (arithmetic costs, memory, parallelism, communication, ...)
 - often depends on architecture (e.g. laptop, GPU clusters, multi-node HPC, ...), all of which I'm experienced with.
- I will develop error estimation and parameter selection methods

I really enjoy thinking about how to think about technical subjects. Towards this end, I've worked on several “handbooks”:

RandNLA w. Examples

- Online book providing a first introduction to RandNLA, with many accompanying numerical examples.

The Lanczos algorithm for matrix functions: a handbook for scientists

- Accessible introduction to Lanczos-based methods for matrix functions, with a particular emphasis on conceptual understanding of the algorithm in finite precision arithmetic.

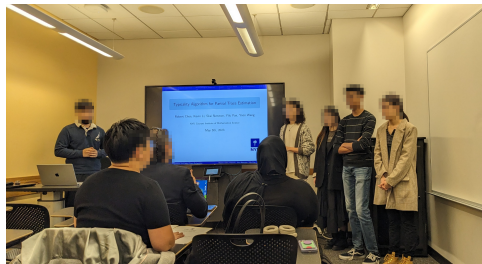
³links at <https://research.chen.pw>

Student Research

I've supervised over a dozen independent studies and research projects, including three which lead to publications.⁴

Students were able to present their work:

- Temple Numerical Analysis Day
- SIAM NY-NJ-PA Annual Meeting
- NYU Undergraduate Research Conference
- Alan Edelman's 60th B-day Conference



I'm very excited to continue this work with students at Shanghai Tech!

⁴Xu and Chen 2024; Chen, Chen, et al. 2024; Chen, Huber, Lin, and Zaid 2026.

Thank You



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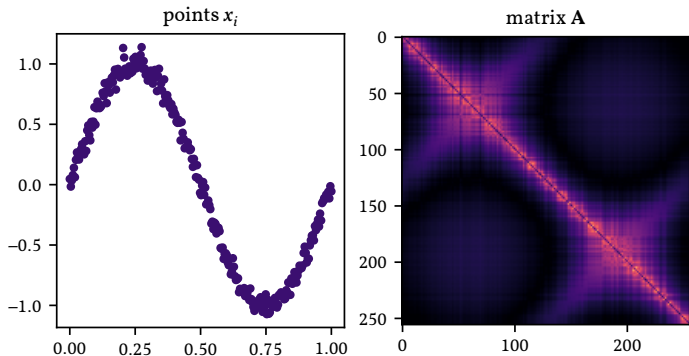
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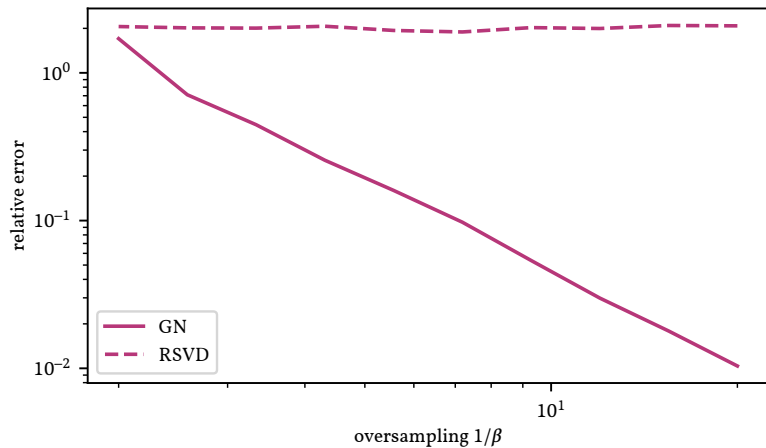
Bonus slides

Numerical experiment

Given points $x_i \in \mathbb{R}^2$, define $[\mathbf{A}]_{i,j} = -\log(\|x_i - x_j\|)$



Numerical experiment



The RSVD tries to compute $\mathbf{Q}^\top \mathbf{B}$ directly; this is the solution to:

$$\min_{\mathbf{X}} \|\mathbf{A} - \mathbf{Q}\mathbf{X}\|_F.$$

Instead, we can solve a sketched problem:

$$\min_{\mathbf{X}} \|\Psi^\top \mathbf{A} - \Psi^\top \mathbf{Q}\mathbf{X}\|_F.$$

This means $\mathbf{X} = (\Psi^\top \mathbf{Q})^\dagger \Psi^\top \mathbf{A}$.

Observation. By adding columns to Ψ , we can damp errors in the product $\Psi^\top \mathbf{A}$.

The algorithm is also non-adaptive (we can do products with Ψ in advance)

⁵Clarkson and Woodruff 2009; Tropp, Yurtsever, Udell, and Cevher 2017; Nakatsukasa 2020.

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⁵Clarkson and Woodruff 2009; Tropp, Yurtsever, Udell, and Cevher 2017; Nakatsukasa 2020.

The RSVD tries to compute $\mathbf{Q}^\top \mathbf{B}$ directly; this is the solution to:

$$\min_{\mathbf{X}} \|\mathbf{A} - \mathbf{Q}\mathbf{X}\|_F.$$

Instead, we can solve a sketched problem:

$$\min_{\mathbf{X}} \|\Psi^\top \mathbf{A} - \Psi^\top \mathbf{Q}\mathbf{X}\|_F.$$

This means $\mathbf{X} = (\Psi^\top \mathbf{Q})^\dagger \Psi^\top \mathbf{A}$.

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Generalized Nyström (perturbation) analysis

Extend $\mathbf{Q}$ to an orthogonal matrix $[\mathbf{Q} \ \widehat{\mathbf{Q}}]$, and write $\boldsymbol{\Psi}_1 = \boldsymbol{\Psi}^\top \mathbf{Q}$ and $\boldsymbol{\Psi}_2 = \boldsymbol{\Psi}^\top \widehat{\mathbf{Q}}$.

By orthogonal invariance, $\boldsymbol{\Psi}_1$ and $\boldsymbol{\Psi}_2$ are independent Gaussian matrices!

First observe:

$$\boldsymbol{\Psi}^\top \mathbf{B} = \boldsymbol{\Psi}^\top (\mathbf{Q}\mathbf{Q}^\top + \widehat{\mathbf{Q}}\widehat{\mathbf{Q}}^\top) \mathbf{B} = \boldsymbol{\Psi}_1 \mathbf{Q}^\top \mathbf{B} + \boldsymbol{\Psi}_2 \widehat{\mathbf{Q}}^\top \mathbf{B}.$$

Therefore:

$$\mathbf{X} = (\boldsymbol{\Psi}^\top \mathbf{Q})^\dagger (\boldsymbol{\Psi}^\top \mathbf{B}) = \boldsymbol{\Psi}_1^\dagger \boldsymbol{\Psi}_1 \mathbf{Q}^\top \mathbf{B} + \boldsymbol{\Psi}_1^\dagger \boldsymbol{\Psi}_2 \widehat{\mathbf{Q}}^\top \mathbf{B} = \mathbf{Q}^\top \mathbf{B} + \boldsymbol{\Psi}_1^\dagger \boldsymbol{\Psi}_2 \widehat{\mathbf{Q}}^\top \mathbf{B}.$$

Adding more columns to $\boldsymbol{\Psi}$ (and hence $\boldsymbol{\Psi}_1$) reduces the error in the second term.

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Extend $\mathbf{Q}$ to an orthogonal matrix $[\mathbf{Q} \ \widehat{\mathbf{Q}}]$, and write $\boldsymbol{\Psi}_1 = \boldsymbol{\Psi}^\top \mathbf{Q}$ and $\boldsymbol{\Psi}_2 = \boldsymbol{\Psi}^\top \widehat{\mathbf{Q}}$.

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First observe:

$$\boldsymbol{\Psi}^\top \mathbf{B} + \mathbf{E} = \boldsymbol{\Psi}^\top (\mathbf{Q}\mathbf{Q}^\top + \widehat{\mathbf{Q}}\widehat{\mathbf{Q}}^\top) \mathbf{B} + \mathbf{E} = \boldsymbol{\Psi}_1 \mathbf{Q}^\top \mathbf{B} + \boldsymbol{\Psi}_2 \widehat{\mathbf{Q}}^\top \mathbf{B} + \mathbf{E}.$$

Therefore:

$$\mathbf{X} = (\boldsymbol{\Psi}^\top \mathbf{Q})^\dagger (\boldsymbol{\Psi}^\top \mathbf{B} + \mathbf{E}) = \boldsymbol{\Psi}_1^\dagger \boldsymbol{\Psi}_1 \mathbf{Q}^\top \mathbf{B} + \boldsymbol{\Psi}_1^\dagger \boldsymbol{\Psi}_2 \widehat{\mathbf{Q}}^\top \mathbf{B} + \boldsymbol{\Psi}_1^\dagger \mathbf{E} = \mathbf{Q}^\top \mathbf{B} + \boldsymbol{\Psi}_1^\dagger \boldsymbol{\Psi}_2 \widehat{\mathbf{Q}}^\top \mathbf{B} + \boldsymbol{\Psi}_1^\dagger \mathbf{E}.$$

Adding more columns to $\boldsymbol{\Psi}$ (and hence $\boldsymbol{\Psi}_1$) reduces the error in the second term.

Lower bounds

Our algorithm produces a $(1 + \varepsilon)$ -optimal approximation using $O(k \text{ poly}(\log(n)/\varepsilon))$ matvecs.

This is optimal:

Theorem. There is a constant $C > 0$ such that for any k, n, ε , there exists a (random) matrix $\mathbf{A}$ such that getting a $(1 + \varepsilon)$ -optimal HODLR approximation requires at least $C(k \log_2(n/k) + k/\varepsilon)$ matvecs.

Hidden Random Matrix Technique

Our lower bound makes use of a lower-bound I developed in Amsel, Chen, et al. 2024 for *fixed-sparsity* approximation.

Proposition. Fix any sparsity pattern with $\Theta(s)$ nonzero entries in each row and column. There exists a matrix $\mathbf{A}$ such that getting a $(1 + \varepsilon)$ approximation of this sparsity requires at least $\Omega(s/\varepsilon)$ matvecs.

Our approach builds on the increasingly popular **hidden random matrix** technique⁶.

⁶Simchowitz, El Alaoui, and Recht 2018; Braverman, Hazan, Simchowitz, and Woodworth 2020; Dereziński, Epperly, and Meyer 2026.

Intuition

Let $\mathbf{a}$ be a length- d Gaussian vector. Consider what happens after we observe $\mathbf{a}^\top \mathbf{x}$ (wlog $\|\mathbf{x}\| = 1$).

Let $\mathbf{X} \in \mathbb{R}^{d \times (d-1)}$ be an orthonormal basis orthogonal to $\mathbf{x}$. Then $\begin{bmatrix} \mathbf{x} & \mathbf{X} \end{bmatrix}$ is orthogonal. Hence, by the **orthogonal invariance** of the Gaussian distribution

$$\mathbf{a}^\top \begin{bmatrix} \mathbf{x} & \mathbf{X} \end{bmatrix} = \begin{bmatrix} \mathbf{a}^\top \mathbf{x} & \mathbf{a}^\top \mathbf{X} \end{bmatrix}$$

is a length- d Gaussian vector.

We observe $\mathbf{a}^\top \mathbf{x}$, but don't know anything about $\mathbf{a}^\top \mathbf{X} \in \mathbb{R}^{d-1}$.

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